MINDEASE: A WEB-BASED MENTAL WELLNESS APPLICATION USING CBT AND AI

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Project Synopsis

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1. Introduction

Mind Ease is a web-based mental wellness application that helps users manage stress, anxiety, and negative thoughts. It combines CBT techniques, mood tracking, calming exercises, and an AI companion called USHA in one platform.

It includes a CBT Thought Diary, mood tracking, 4-7-8 breathing, 5-4-3-2-1 grounding, and crisis detection. Mind Ease is designed as a self-help tool and not a replacement for professional care.

2. Objectives of the Project

- To provide a guided CBT thought diary that helps users identify and reframe distorted thoughts.
- To track daily mood alongside lifestyle factors and reveal meaningful correlations over time.
- To offer conversational and somatic support for moments of stress through an AI companion and guided calming exercises.
- To protect sensitive personal data through strong encryption, secure authentication and regulatory-compliant data handling

3. Problem Statement

Existing approaches to everyday mental wellness leave several gaps that this project seeks to address:

- Cognitive distortions: Individuals under stress frequently experience automatic negative thoughts, such as catastrophizing or mind-reading, which are difficult to recognize and correct without structured guidance.

- Limited access to care: Professional therapy is often expensive, involves long waiting lists, and is rarely available now of acute need.

- Privacy concerns: Many wellness applications monetize user data, which discourages honest self-disclosure in a highly sensitive context.

4. Scope of the Project

Mood Tracker

Users can give their mood a score from 1 to 10. They can also record things like sleep, exercise, and screen time.

USHA AI Companion

USHA is an AI chatbot that talks with users and provides basic emotional support using Google Gemini API.

Calming Tools

The application provides simple exercises like 4-7-8 breathing and 5-4-3-2-1 grounding to help users feel calm.

Crisis Detection

The system checks for dangerous or concerning words in user messages. If detected, it shows important help numbers like 988, Tele-MANAS, and 112.

Privacy and Security

The application keeps user information safe using encryption, secure passwords, two-factor authentication, and account protection.

5. Methodology

The frontend is developed using React, TypeScript, and Tailwind CSS. The backend uses Node.js and Express. PostgreSQL is used to store the data.

Step 4 – Security Implementation

The system protects user data using secure passwords, encryption, and two-factor authentication.

Step 5 – Testing

The application is tested to make sure login, data security, crisis detection, and other features work correctly.

Step 6 – Implementation

After testing, all the features are combined into one web application for users to access easily.

6. Hardware and Software Requirements

Hardware Requirements:

Processor: Intel i3 or above

• RAM: 4 GB minimum

• Hard Disk/Storage: 10 GB or more available space

• Internet Connection: Required for AI API and online services

Software Requirements:

• Operating System: Windows 11

• Frontend: React 18, TypeScript

• Backend: Node.js, Express

• Development Environment: Visual Studio Code

• Database: PostgreSQL

• Database Hosting: Supabase

• AI Integration: Google Gemini API

• Security: Bcrypt, AES-256, TOTP-based 2FA

• Web Browser: Google Chrome or any modern web browser

7. Expected Outcome

Mind Ease is expected to provide a simple web-based platform that:

- Helps users manage negative thoughts and track their mood.
- Provides AI support through USHA.
- Offers breathing and grounding exercises.
- Detects concerning inputs and provides crisis-support resources.
- Protects user data through secure authentication and encryption.

Overall, the project aims to provide an accessible and privacy-conscious mental wellness support platform.

8. Conclusion

Mind Ease combines CBT techniques, mood tracking, AI support, calming exercises, and security features in one web application. It aims to help users manage everyday stress and negative thoughts while protecting their personal information.

It is designed as a self-help tool, not a replacement for professional mental health care.

9. References